

AI & ML Engineering

Abdullah Saihan

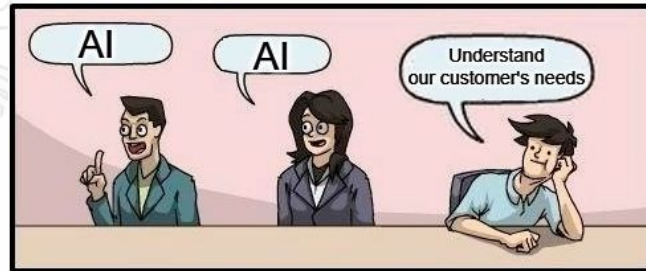
Artificial Intelligence Engineer

Who am I



- I have done my undergrad in Mechatronics Engineering from KUET
- Currently I'm working as an AI Engineer at Banglalink
- Previously, I worked as a Machine Learning Engineer at multiple companies
- I have also worked as an IoT and Embedded Systems Engineer
- Provided mentorship and training in programming and robotics



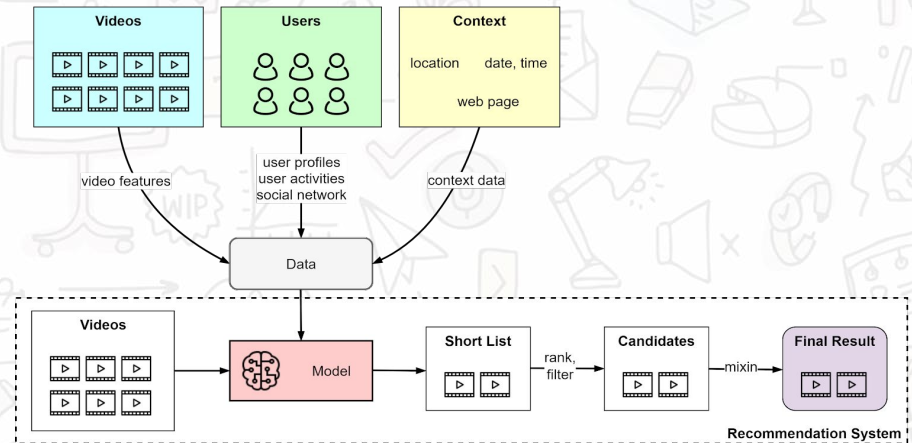


What is Artificial Intelligence (AI) ?

AI is when machines are trained to think or act like humans ...

- YouTube recommending videos
- Google Maps finding fastest route
- Face unlock on phone
- Google Translate
- Voice Assistant on phone

AI basically learns pattern from data.







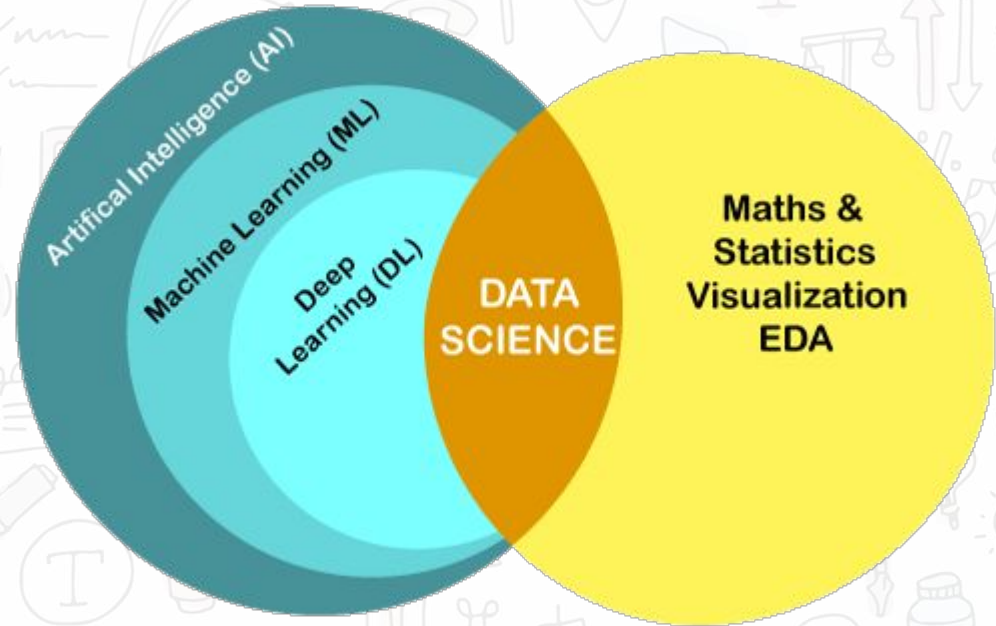


Data Science Vs AI/ML



→ Data science is about turning **raw data into useful knowledge**.

→ Artificial intelligence teaches computers to act and decide like humans



The AI landscape



Artificial Intelligence

AI involves techniques that equip computers to emulate human behavior, enabling them to learn, make decisions, recognize patterns, and solve complex problems in a manner akin to human intelligence.

Machine Learning

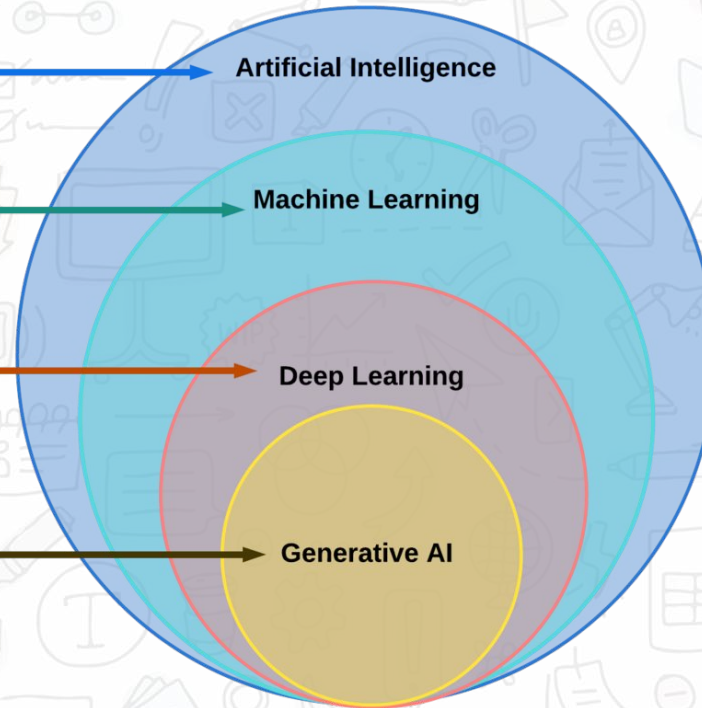
ML is a subset of AI, uses advanced algorithms to detect patterns in large data sets, allowing machines to learn and adapt. ML algorithms use supervised or unsupervised learning methods.

Deep Learning

DL is a subset of ML which uses neural networks for in-depth data processing and analytical tasks. DL leverages multiple layers of artificial neural networks to extract high-level features from raw input data, simulating the way human brains perceive and understand the world.

Generative AI

Generative AI is a subset of DL models that generates content like text, images, or code based on provided input. Trained on vast data sets, these models detect patterns and create outputs without explicit instruction, using a mix of supervised and unsupervised learning.

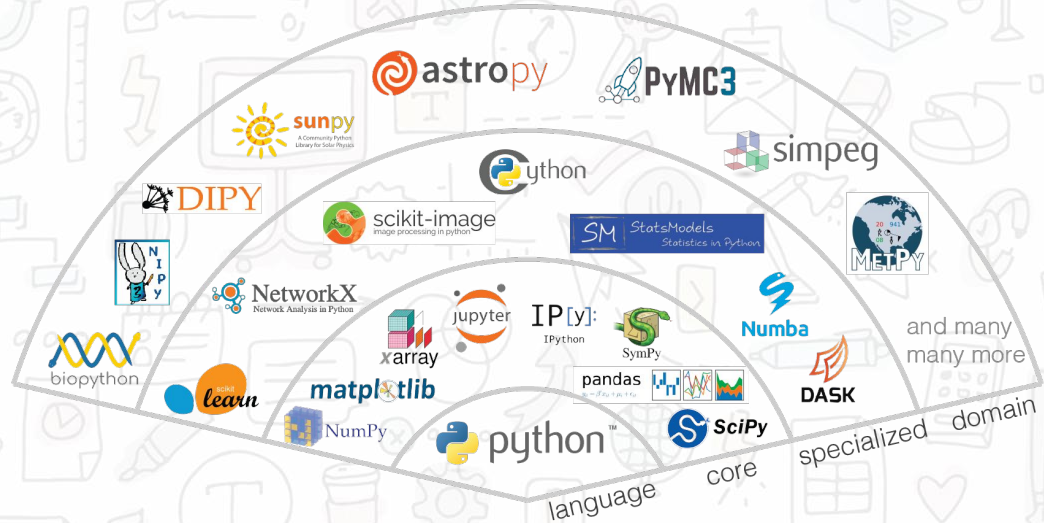


1. Programming Foundations: Learn Python



Why Python?

- Most AI tools use Python
- Beginner friendly
- Huge ecosystem



2. Learn Data Handling

AI requires working with data.

Learn:

- NumPy
- Pandas
- Data cleaning
- Data Visualization

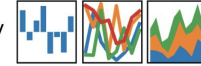
Data
Manipulation
Library

pandas

$$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$$



xarray



Visualisation
Library

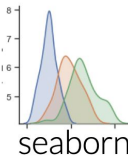
matplotlib



bokeh



ggplot2



seaborn

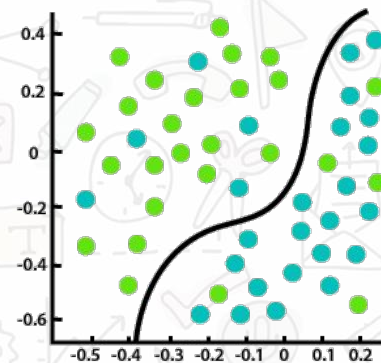
3. Understand Machine Learning Basics



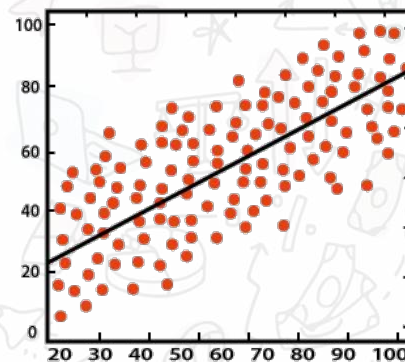
Machine learning is a subset of **artificial intelligence** that enables computer systems to **learn from data**.

Understand concepts:

- Supervised and Unsupervised learning
- Classification and Regression
- Overfitting / Underfitting
- Train/Test split
- Evaluation metrics (Accuracy, F1 etc)



Classification



Regression



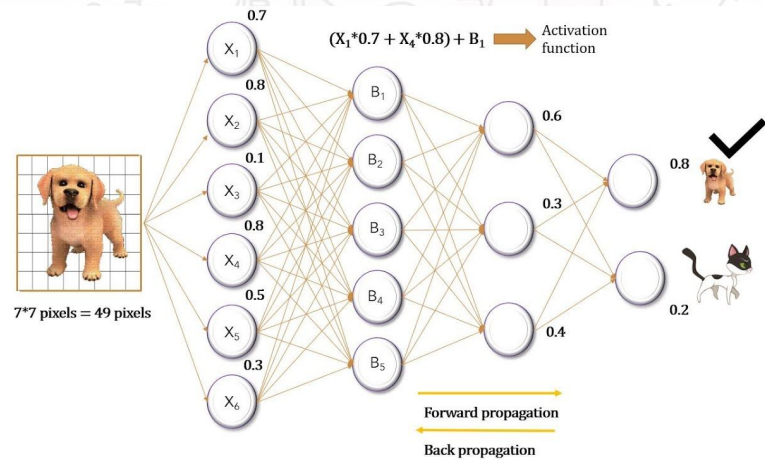
4. Understanding Deep Learning Basics



Deep Learning is a **subset of ML** that uses **neural networks with many layers** to learn patterns from large amounts of data.

Understand concepts:

- Neural Networks
- Layers and Activations
- Gradient Descent (How model learns)
- Backward and forward propagations



TensorFlow



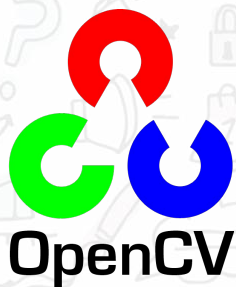
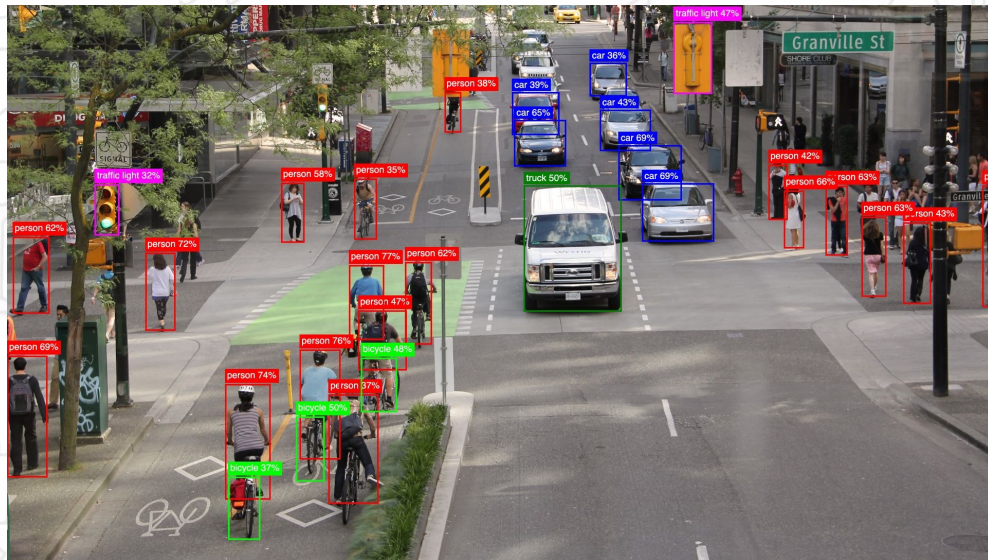
PyTorch

5. Understanding Computer Vision



Key concepts:

- Image Processing
- Convolutional Neural Network
- Image Classification
- Image Segmentation
- Object Detection



ultralytics

Computer Vision



Is this a dog?



Image Classification

What is there in image
and where?



Object Detection

Which pixels belong to
which object?

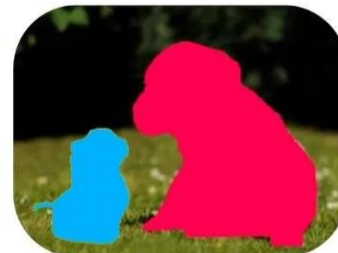
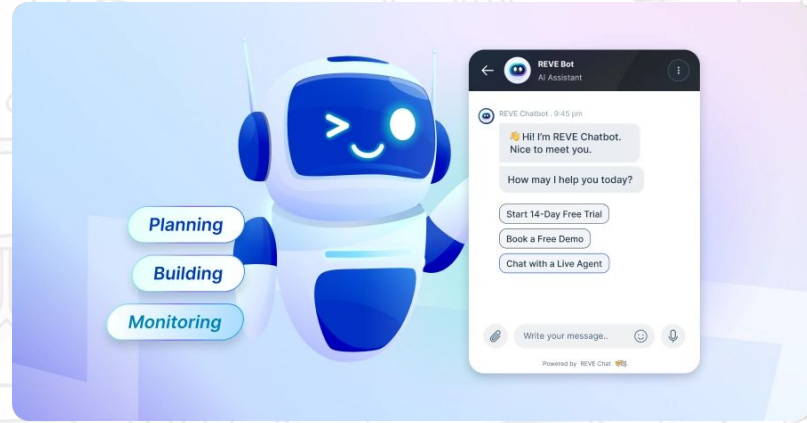


Image Segmentation

6. Natural Language Processing Basics

Key Concepts:

- Text Processing
- Language Models
- RNN and Transformers
- Text Embedding
- Sentiment Analysis
- Text Generation

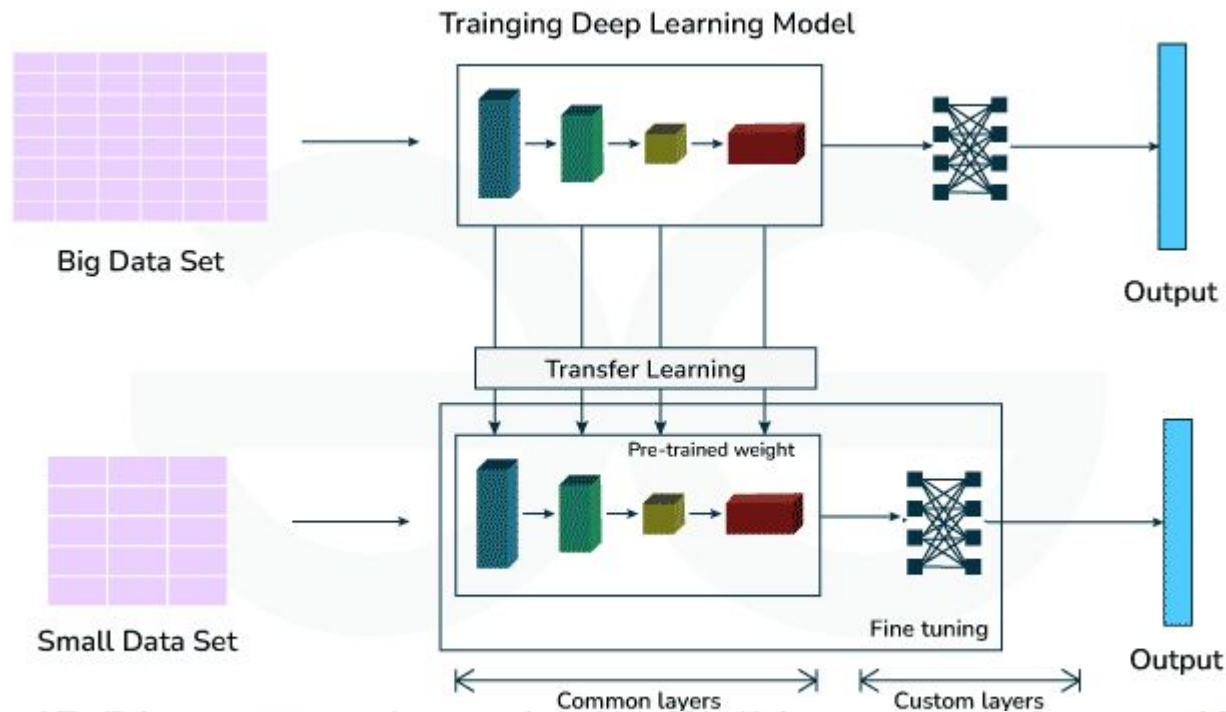


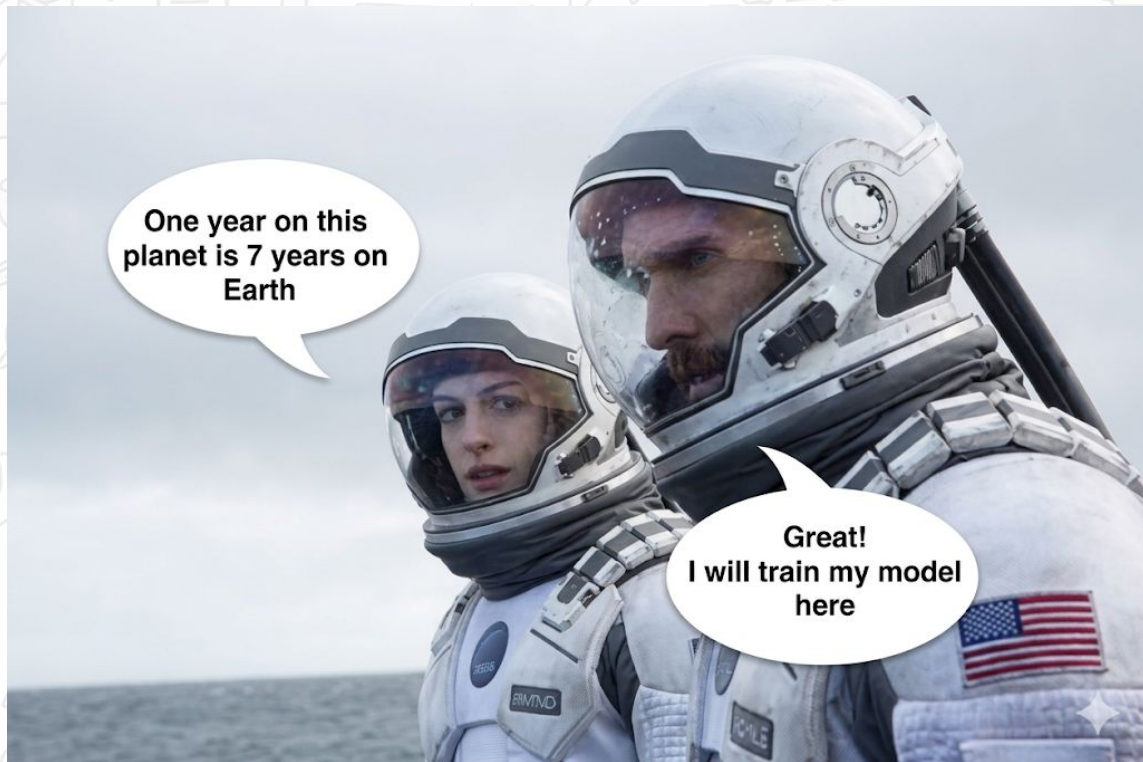
spaCy



Hugging Face

Model Fine Tuning





7. Learn Modern AI (LLMs)

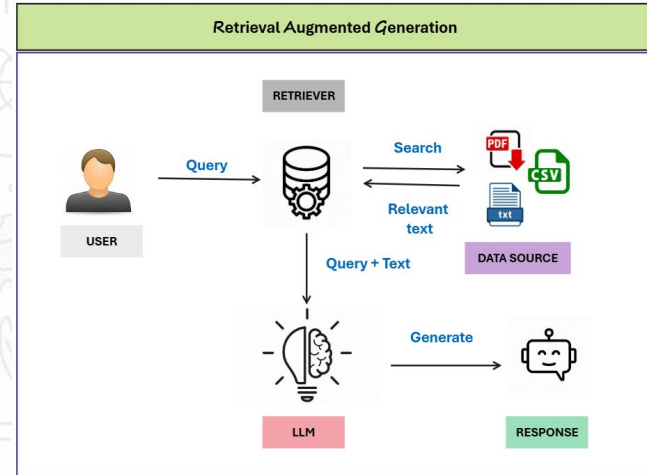
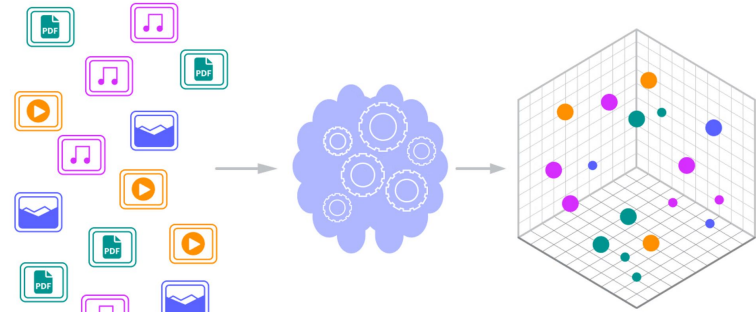
Today AI engineering heavily involves:

Large Language Models

Examples: GPT, Gemini, Claude, Llama, Qwen

Learn:

- Prompt engineering (React, Structured O/P)
- API usage
- Embeddings
- Vector databases
- RAG (Retrieval Augmented Generation)



How LLM Works

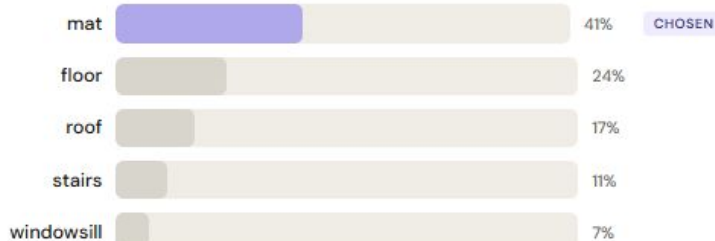


How LLM Works

STEP 1 — PREDICTING WORD 1

The cat sat on the **mat**

TOP CANDIDATES CONSIDERED



✓ Chosen: "mat" — "mat" wins — it learned this phrase from millions of books and websites. "The cat sat on the mat" is very common in English.

→ Predict next word

↶ Start over



Probabilities, not certainty

The model scores every word in its vocabulary (50,000+) and picks the most likely one based on context.



Learned from text

These probabilities come from training on billions of books, articles, and websites — it learned what words follow other words.



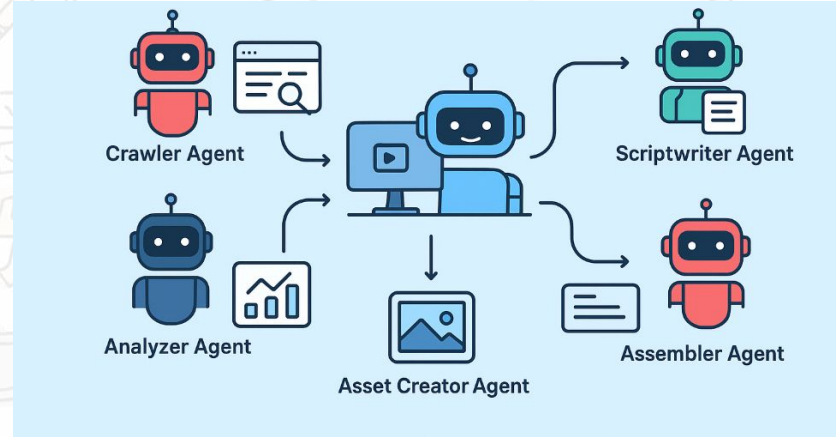
Autoregression

Each new word is fed back as input. A 200-word reply means running this prediction loop 200 times.

8. Working with Agents

Key Concepts:

- Autonomy
- Function Calling
- Tool Usage
- Reasoning & Planning
- Model Context Protocol



LlamaIndex

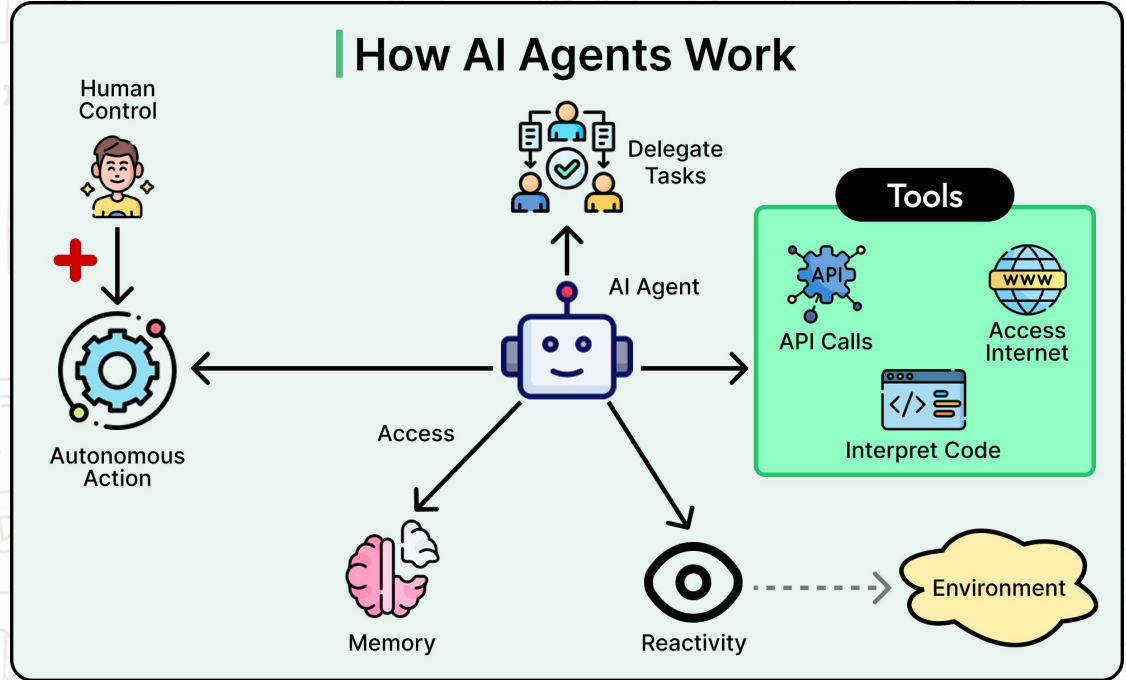


LangGraph

Working with Agents

If **LLM = brain** - Thinks, reasons, generates text, But **cannot act on its own**

Then, **Agent = brain + body + tools + decision-making loop**



8. MLOps & Production Engineering

Employers hire engineers who can **ship**
reliable AI Systems, not just notebooks..

Understand concepts:

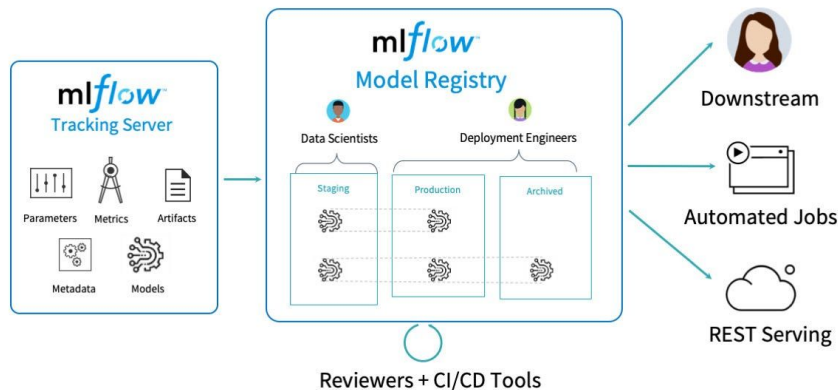
- Fastapi
- Git and Github/Gitlab
- Docker
- Model serving, and monitoring



docker



Ollama



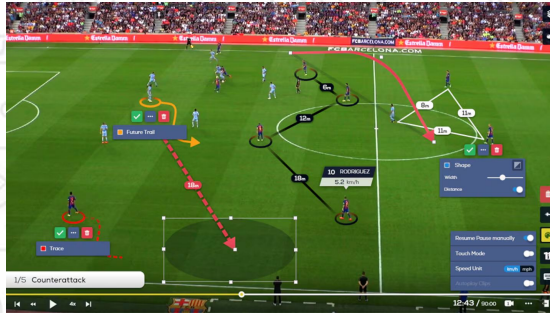
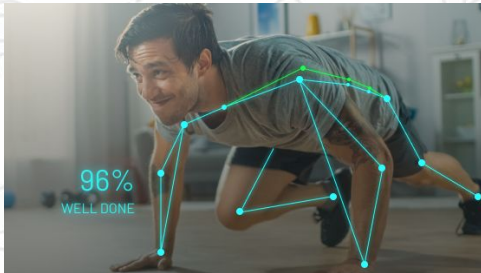
Build Real Projects



Projects are **more important than certificates.**

Examples:

- Customer Support Chatbot
- AI Football Match Analyzer
- AI Marketing Agent
- AI Fitness Coach



Intro to Machine Learning

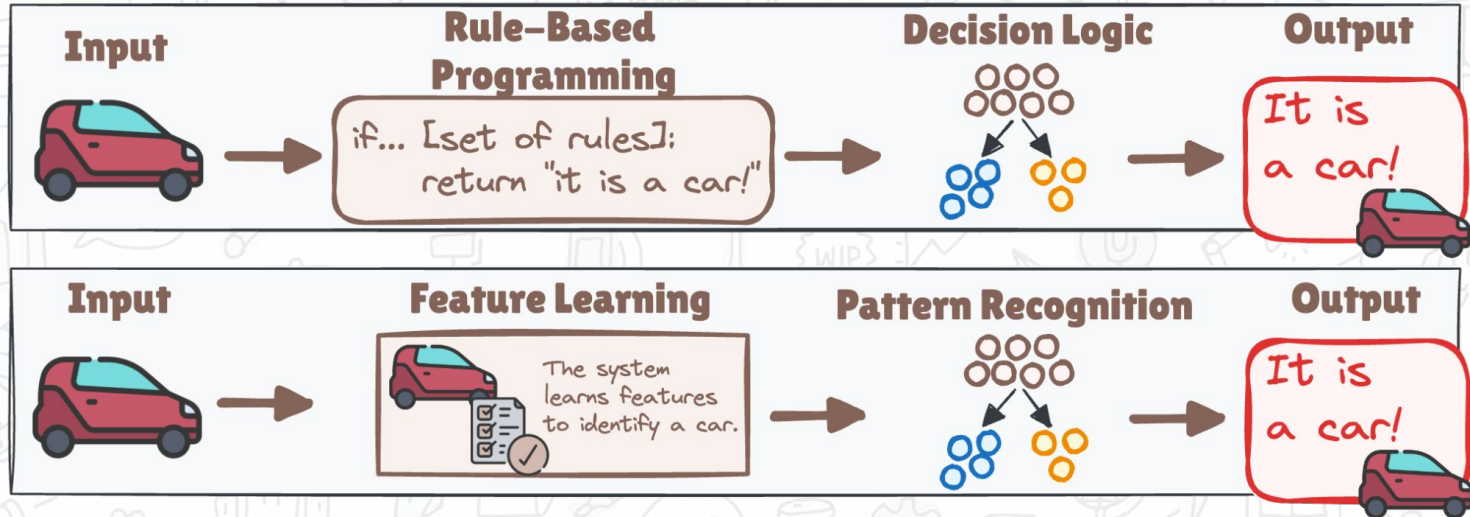
What is Machine Learning?

Machine learning is a subset of artificial intelligence (AI) that enables systems to **learn from data**, **identify patterns**, and **make decisions or predictions** without being explicitly programmed for every specific rule

Think of it like this: instead of giving the computer exact instructions, you **show it examples**, and it figures out the patterns on its own.

Rule-based vs ML System

Rule-based vs ML System

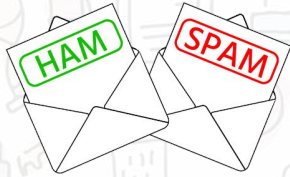


Rule Based vs ML System Simulation

Type of Machine Learning

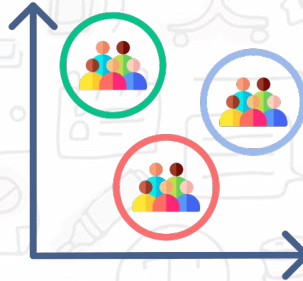
Machine Learning

Supervised Learning



Labels required

Unsupervised Learning



No labels required

Reinforcement Learning



Learn from mistakes

Labelled vs Unlabelled Data



Labelled data



Labelled data



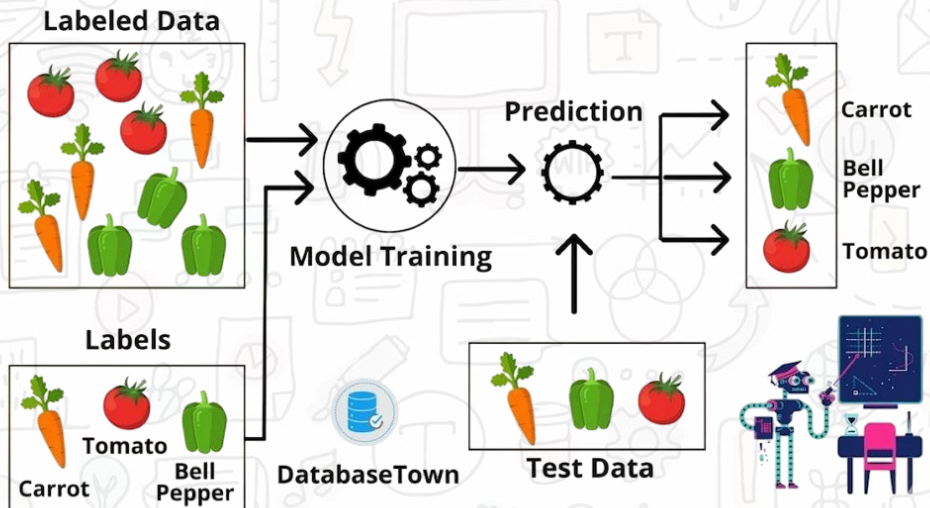
Unlabelled data



Supervised Learning

SUPERVISED LEARNING

Supervised machine learning is a branch of artificial intelligence that focuses on training models to make predictions or decisions based on labeled training data.



Supervised Learning



Supervised Learning Applications:

- **Email spam detection** – Classifying emails as spam or not spam.
- **Predicting house prices** – Using features like size, location, and rooms to estimate price.
- **Medical diagnosis** – Predicting diseases from patient symptoms or test results.
- **Credit scoring** – Deciding whether a person should get a loan based on financial history.

KEY APPLICATIONS OF SUPERVISED LEARNING



EMAIL FILTERING

Classifying emails as
spam or not spam



MEDICAL DIAGNOSIS

Predicting diseases
based on symptoms



FINANCIAL ANALYSIS

Predicting
stock prices



IMAGE RECOGNITION

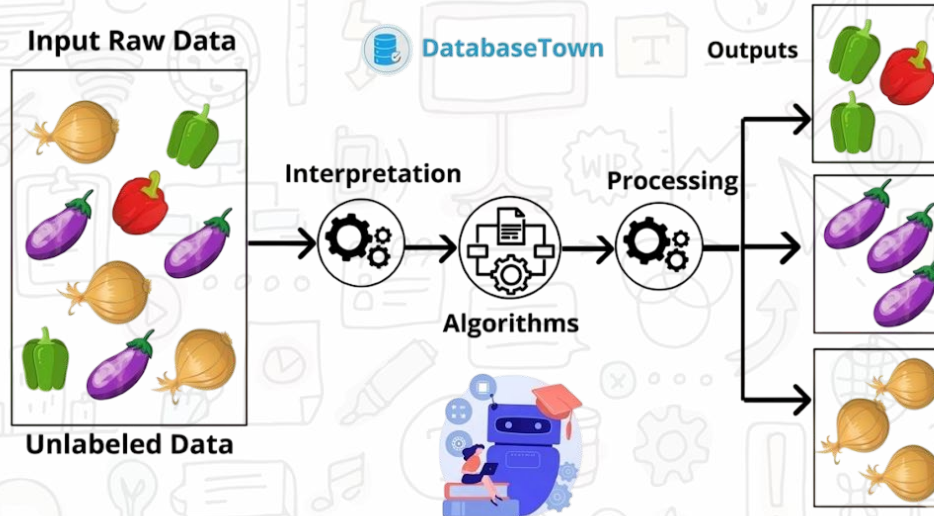
Identifying objects
within images

Unsupervised Learning



UNSUPERVISED LEARNING

Unsupervised learning is a type of machine learning where the algorithm learns from unlabeled data without any predefined outputs or target variables.



Unsupervised Learning

Unsupervised Learning Applications:

- **Customer segmentation** – Grouping customers with similar buying habits for marketing.
- **Market basket analysis** – Finding products that are often bought together (association rules).
- **Anomaly detection** – Detecting unusual patterns, like fraud in transactions.
- **Document clustering** – Organizing news articles or documents into topic-based groups.

KEY APPLICATIONS OF UNSUPERVISED LEARNING



MARKET BASKET ANALYSIS

Discovering products that are often purchased together



GENETIC CLUSTERING

Grouping genes with similar expression patterns



SOCIAL NETWORK ANALYSIS

Identifying communities within large networks



ANOMALY DETECTION

Spotting fraudulent transactions in banking

Supervised vs Unsupervised



supervised learning

Input data



Annotations

These are
apples



Model



Prediction

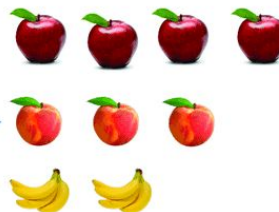
Its an
apple!

unsupervised learning

Input data



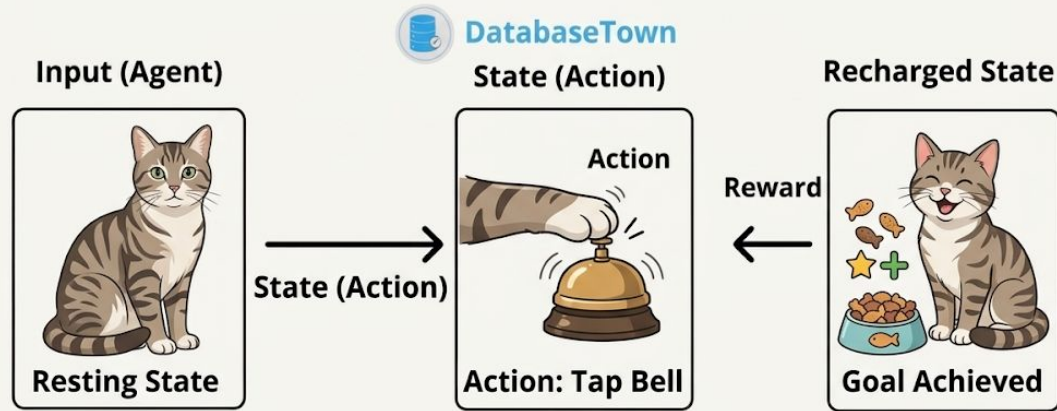
Model



Reinforcement Learning

REINFORCEMENT LEARNING

Reinforcement learning is a machine learning paradigm that focuses on how agents learn to interact with an environment to maximize cumulative rewards.



Algorithms and Approaches in Reinforcement Learning

- Q-learning
- Deep Q-networks (DQN)
- Policy Gradients Methods
- Proximal Policy Optimization (PPO)

Reinforcement Learning



Reinforcement Learning Applications:

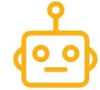
- **Game playing** – Teaching AI to play games like **Chess, Go, or video games** (e.g., AlphaGo).
- **Robotics** – Teaching robots to **walk, pick objects, or navigate** in complex environments.
- **Recommendation systems** – Optimizing recommendations dynamically based on **user interactions**.
- **Resource management** – Optimizing **energy usage or traffic signals** for efficiency.

KEY APPLICATIONS OF REINFORCEMENT LEARNING



VIDEO GAMES

Teaching AI to play and excel at complex video games



ROBOTICS

Enabling robots to learn tasks like walking or grasping objects



AUTONOMOUS VEHICLES

Developing systems for self-driving cars to make decisions in real traffic



PERSONALIZED RECOMMENDATIONS

Tailoring suggestions to individual users' preferences over time

Teaching a Robot how to Walk



Type of Supervised Learning



TYPE OF SUPERVISED LEARNING

Classification

Categorical Label
eg. Apple/Orange recognition

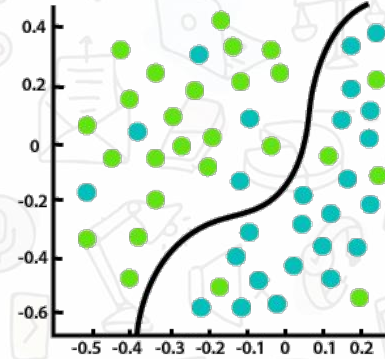
Regression

Numeric Label
eg. House Price Prediction

Classification

Classification

- **Goal:** Predict a **category or class**.
- **Applications:**
 - Email spam detection (spam or not spam).
 - Diagnosing diseases (disease A, disease B, or healthy).
 - Sentiment analysis (positive, negative, neutral).
 - Handwriting or image recognition (digits, letters, objects).

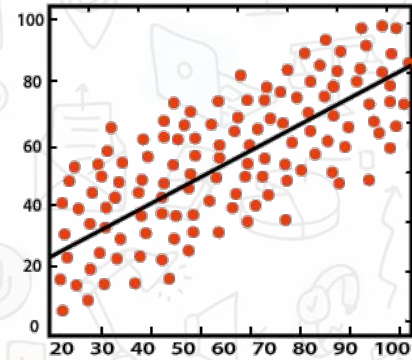


Classification

Regression

Regression

- **Goal:** Predict a **continuous number**.
- **Applications:**
 - Predicting house prices based on size, location, and rooms.
 - Forecasting stock prices or sales.
 - Estimating temperature or weather conditions.



Regression

Classification Vs Regression

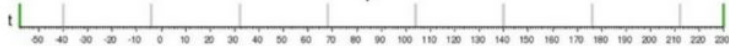


Regression



What will be the temperature tomorrow?

84°



Fahrenheit

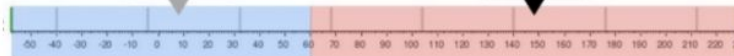
Classification



Will it be hot or cold tomorrow?

COLD

HOT



Fahrenheit

Features and Labels

Features (Input variables)

- **Features** are the **input data** used to make predictions
- They describe the **properties or characteristics** of each example
- Usually denoted as **X**

Example:

If you're predicting house prices:

- Size (sq ft)
 - Number of bedrooms
 - Location
-  These are **features**

Labels (Target / Output)

- **Labels** are the **correct answers** you want the model to learn
- They represent what you're trying to predict
- Usually denoted as **y**

Example:

- House price
 This is the **label**
- **Labels** are also called “**targets**”

Features and Labels

Regression Dataset



Features				Label
Size	Beds	Baths	Zip	Price
1100	1	1	64576	1.29
1900	3	1.5	78321	2.14
2800	3	3	98712	3.10
3400	4	3.5	25721	3.75

Features and Labels

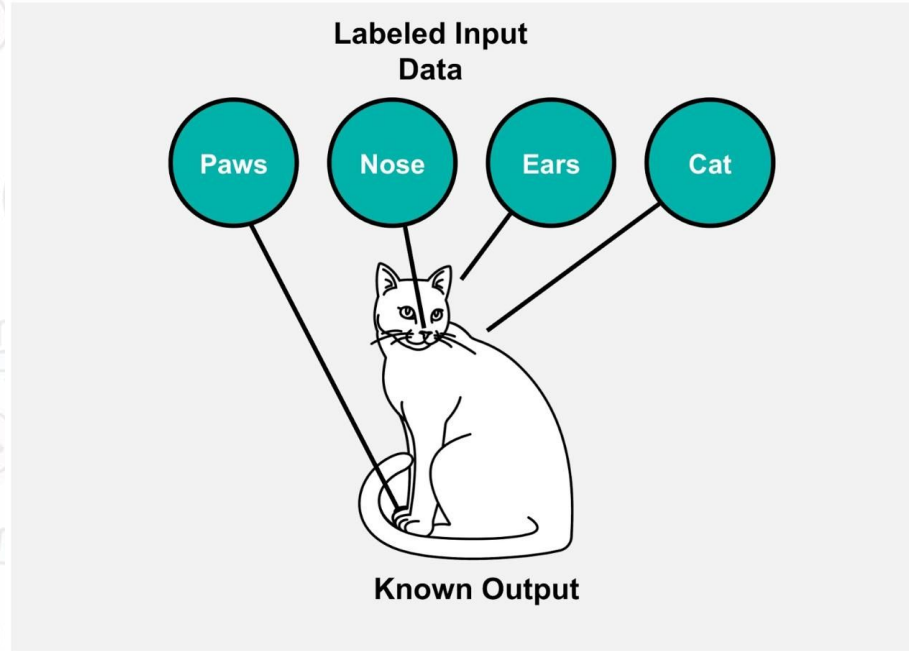


Classification Dataset

Features			label
Sleep Quality	Stress	Weight	Migraine
Poor	Yes	97	Yes
Good	No	78	No
Poor	No	47	Yes
⋮	⋮	⋮	⋮
Moderate	No	76	Yes

Features and Labels

In this image, what
are the features
and what is the
label?



Types of Data

1 Based on Structure

This is about **how data is stored**

- **Structured data** → tables (rows & columns)
- **Unstructured data** → text, images, audio, video
- **Semi-structured data** → JSON, XML

2 Based on Data Type (Values)

This is about **what kind of values the data contains**

- **Numerical**
 - Continuous
 - Discrete
- **Categorical**
 - Nominal
 - Ordinal

Types of Data k21academy

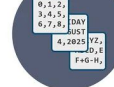
Structured Data vs Unstructured Data

Can be displayed
in rows, columns and
relational databases



XY	1	2
A	A1	A2
B	B1	B2
C	C1	C2
D	D1	D2

Numbers, dates
and strings



9, 3, 2,	
3, 4, 5,	DAY
6, 7, 8,	JUST
4, 2025	TC,
— 0, 8	
F=0-H,	

Estimated 20% of
enterprise data (Gartner)



Requires less storage



Easier to manage
and protect with
legacy solutions



Unstructured Data



Cannot be displayed
in rows, columns and
relational databases



Images, audio, video,
word processing files,
e-mails, spreadsheets



Estimated 80% of
enterprise data (Gartner)



Requires more storage



More difficult to
manage and protect
with legacy solutions

Structured Data



id	name	age
1	Jim	28
2	Pam	26
3	Michael	42

id	subject	Teacher
1	Languages	John Jones
2	Track	Wally West
3	Swimming	Arthur Curry
4	Computers	Victor Stone

student_id	subject_id	grade
2	1	98
1	2	100
1	4	75
3	3	60
2	4	76
3	2	88

Unstructured Data



Text files and documents



Server, website and application logs



Sensor data



Images



Video files



Audio files

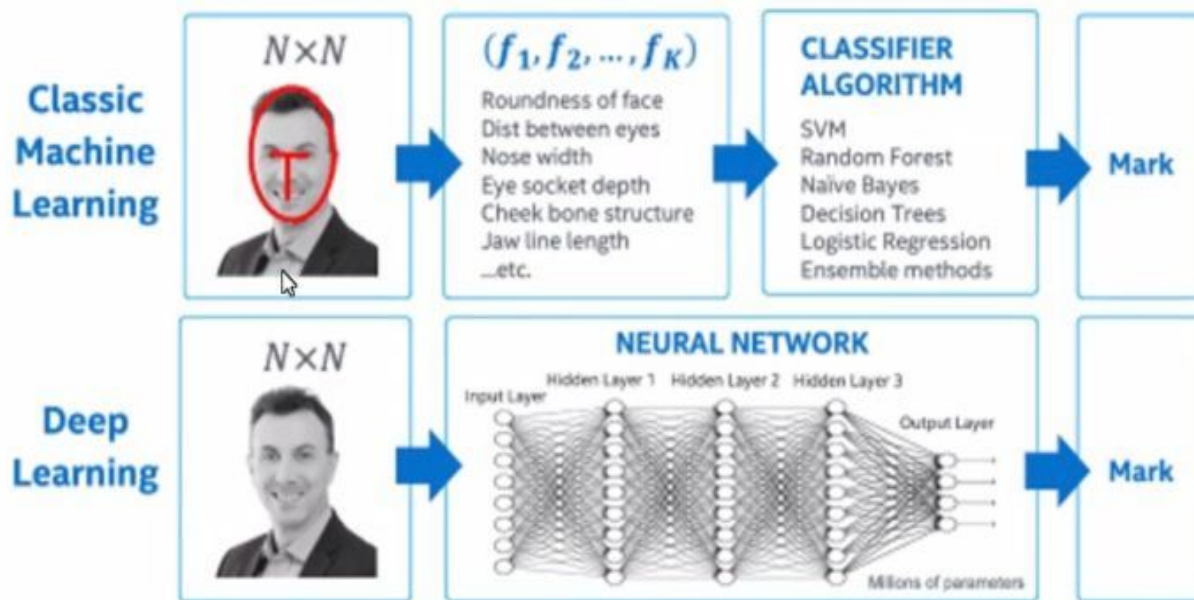


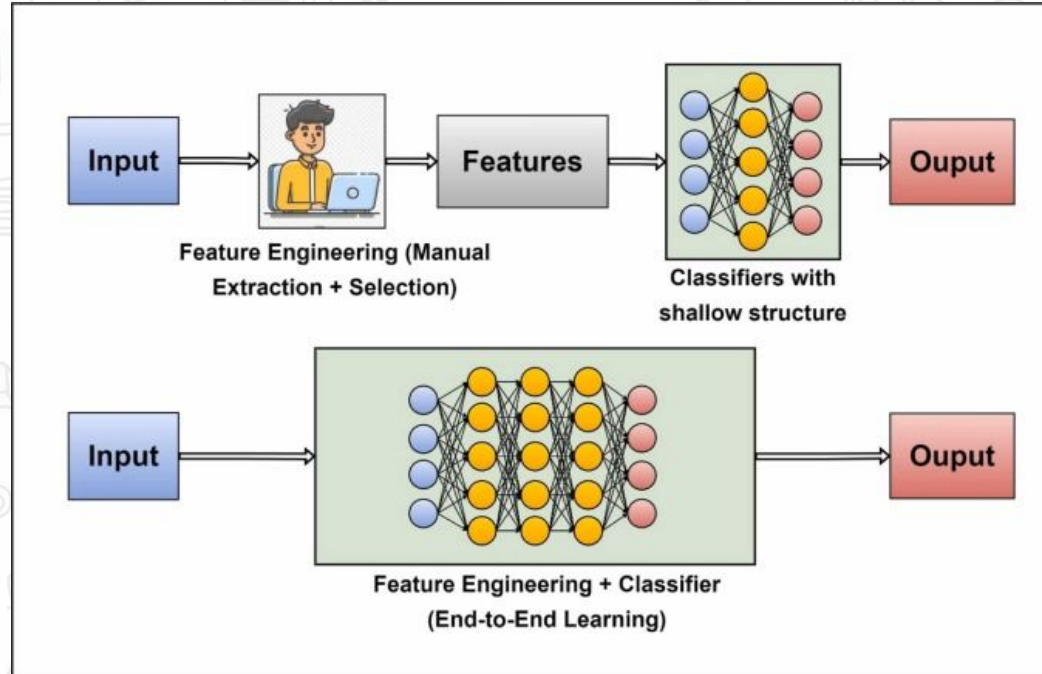
Emails



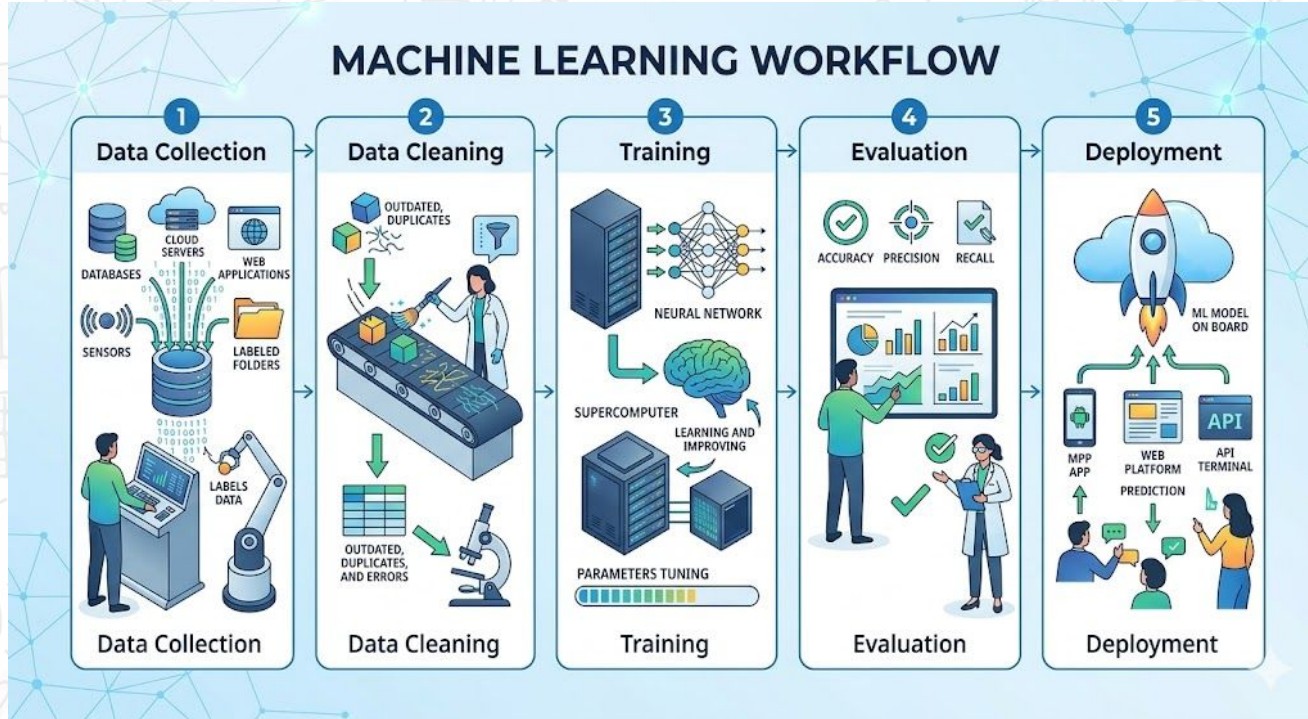
Social media data

Classical ML vs Deep Learning

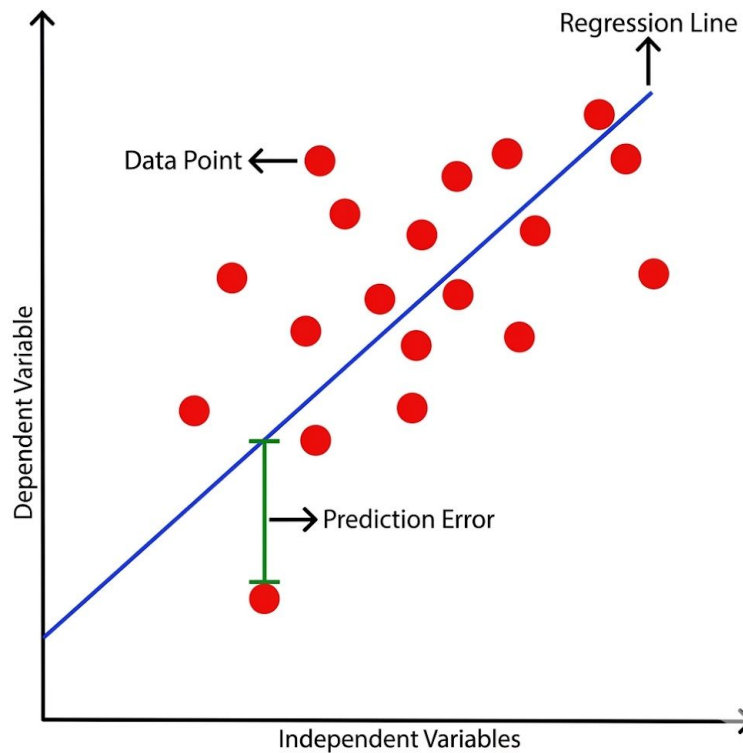




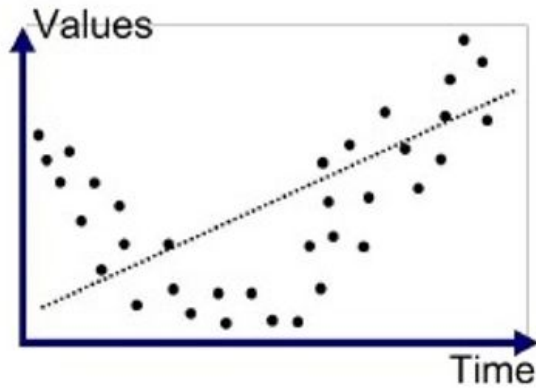
Machine Learning Workflow



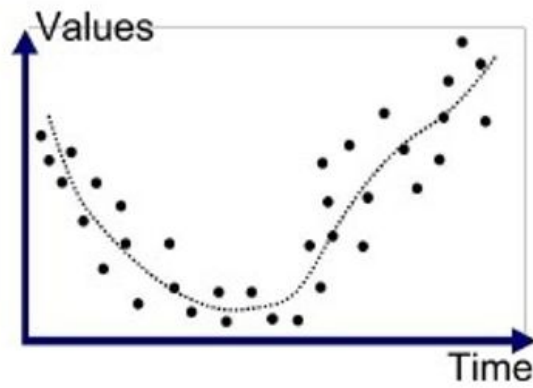
Linear Regression



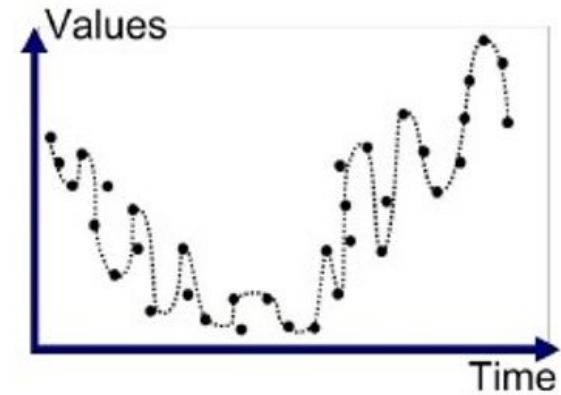
Overfitting vs Underfitting



Underfitted

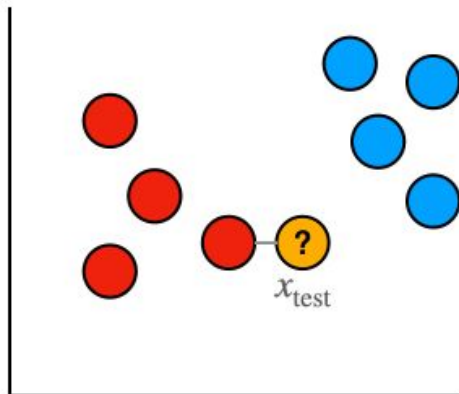


Good Fit/Robust



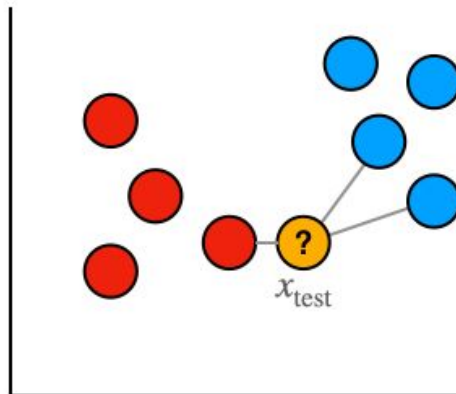
Overfitted

K Nearest Neighbors



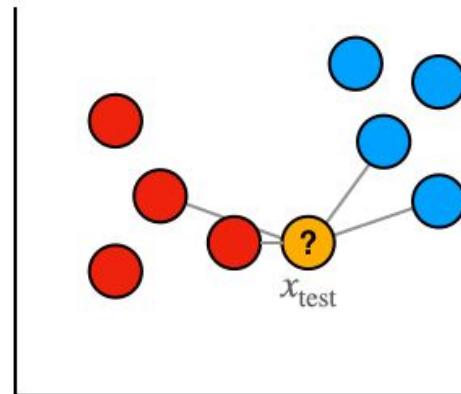
$k = 1$

Nearest point is **red**, so x_{test} classified as **red**



$k = 3$

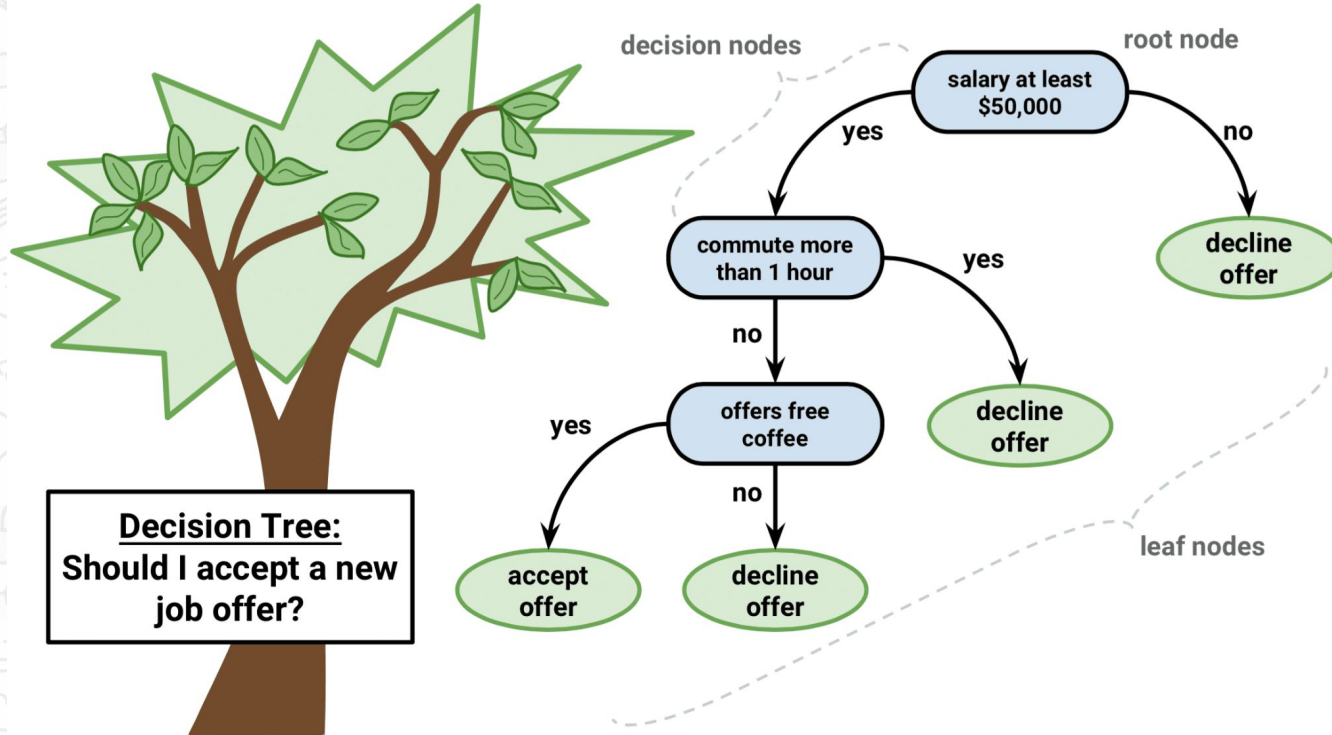
Nearest points are {**red**, **blue**, **blue**} so x_{test} classified as **blue**



$k = 4$

Nearest points are {**red**, **red**, **blue**, **blue**} so classification of x_{test} is not properly defined

Decision Tree



**NO THANKS,
I HAVE
CHATGPT NOW**



Thank you!